Third-Party Cardiometabolic Outcomes in a GLP-1 Telehealth Cohort

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Abstract

Background. Telehealth now delivers most GLP-1 prescribing. These programs measure weight and little else. A member's lab work is ordered by an outside clinician the program never sees. So the only outcomes it can report are self-reported or program-ordered. Those are the two kinds of measurement most open to selection.

Methods. With member consent we linked a U.S. telehealth obesity program to outside clinical records. The link runs through the Carequality and CommonWell health-information exchanges. It returns the laboratory values and vital signs that members' own outside clinicians recorded. So every outcome is measured by a clinician with no relationship to the program. Of 417,307 members with a dispensed GLP-1 and a record link, 49,896 (12.0%) had a measure paired across the first fill. We define disease groups by an outside diagnosis recorded before treatment. We report paired changes. We then decompose each improvement into a weight-mediated part and a part beyond weight, using within-member fixed-effects models.

Results. Across the full diagnosed groups, systolic blood pressure fell 4.85 mmHg among 16,008 members with hypertension. LDL fell 8.55 mg/dL. HbA1c fell 0.57 points in type 2 diabetes. The decomposition attributes 42% of the systolic step and 46% of the HbA1c step to weight, with triglycerides and fasting glucose exploratory, and the remainder arriving beyond weight. The within-member weight slope is attenuated in ways that could bias this share in either direction. So we treat it as uncertain rather than as a bound. This is consistent with SELECT, where early in-trial weight loss was not a significant mediator of semaglutide's cardiovascular benefit [Lincoff et al., 2023, [author list to verify], 2025].

Conclusions. The contribution is a data design. It is a direct-to-consumer telehealth cohort whose outcomes are verified against the patient's own outside clinicians through nationwide health-information exchange. Large clinician-measured GLP-1 cohorts already exist in integrated systems such as the VA and in research networks. What is new here is the linkage from a direct-to-consumer program, which owns only the scale, out to the records those outside clinicians keep. The decomposition is secondary and confirmatory. Its weight share is an estimate whose attenuation could run in either direction. The endpoints are surrogates, not events. The design has no untreated comparison group. These are not causal effect estimates.

1 Introduction

Direct-to-consumer telehealth now delivers most GLP-1 prescribing in the United States. It reports its results in the only unit it owns, the scale. A telehealth program records weight often because

*Mochi Health. This is an observational analysis of a treated population, built on the de-identified aggregate tables in the Covered-with-Mochi evidence package. Every number here reproduces from those tables, which are listed in the data appendix. It is not peer reviewed, not externally audited, and no claim here is approved for external use. The paper reports measured changes in laboratory values. It does not count clinical events, and the outcome figures in Section 5 are projections from published trials, not events observed at Mochi.

it sets the weight. It records blood pressure, blood sugar, and lipids almost never. Those are drawn at a clinic it does not run, by a clinician it does not employ, and they land in a chart it cannot read. Every outcome number a telehealth weight program publishes is therefore either self-reported or ordered by the program itself. Those are the two measurements most exposed to selection and to the program's own interest in the answer. This is the credibility problem at the center of direct-to-consumer obesity care. It is a data problem before it is a clinical one.

The contribution of this paper is a data design that removes that problem. With member consent we linked a U.S. telehealth obesity program to outside clinical records. The link runs through the Carequality and CommonWell health-information exchanges, the two networks that connect most large American health systems. It returns the laboratory values and vital signs that members' own outside clinicians recorded. Every outcome we report is measured by a clinician with no relationship to the program, no stake in the result, and no knowledge that the measurement would be analyzed. Large clinician-measured GLP-1 cohorts already exist inside integrated health systems such as the Veterans Affairs system and in research networks such as Truveta, often with comparison groups and at larger scale. What is new here is the linkage from a direct-to-consumer program, which generates only the scale, out to the outside records. The closest direct-to-consumer telehealth cohorts rely on program-ordered or self-reported measurement. That is the circularity this design breaks. The linkage returned 8.77 million readings across 417,307 members. Of those, 49,896 had a measure paired across the first dispensed fill. The coverage fraction is itself a finding about how usable exchange data now is for telehealth evidence, and we report it.

With outcomes that a skeptic can trust, we ask the question the program's economics turn on. Of the measured improvement, how much travels through weight loss, and how much arrives beyond it? This is a decomposition. We frame it deliberately as a secondary analysis rather than the headline, because the weight-independence of GLP-1 benefit is not new. The SELECT trial already established, at the level of hard cardiovascular events, that semaglutide's benefit was largely independent of weight loss [Lincoff et al., 2023, [author list to verify], 2025]. Our decomposition cannot and does not re-establish that for hard endpoints, which we do not measure. What it adds is where our surrogate biomarkers sit relative to that result, qualitatively. SELECT found early weight loss was not a significant mediator of its hard endpoint, and our decomposition finds a substantial part of each biomarker change arriving beyond weight. We report the agreement in sign and do not draw a quantitative contrast from it, because our share is uncertain in direction. Locating the biomarker figure in a real-world population that SELECT's enrollment criteria exclude is the secondary contribution.

The paper proceeds in that order. We describe the linkage and the cohort. We state the identifying assumptions with a causal diagram. We report the paired changes measured by outside clinicians. We then present the decomposition and set it beside SELECT. Finally we hold the changes against the published trial literature, while keeping those projections strictly separate from what we measured.

2 Data

2.1 Linkage and health-information exchange

Members of a U.S. telehealth obesity program operating in all 50 states were linked, with consent, to outside clinical records. The link runs through the Carequality and CommonWell health-information exchanges, the two networks that connect most large American health systems. When a member consents, the program retrieves the structured results any participating system holds for that person: laboratory panels, vital signs, diagnoses, and medication records. These data are

generated by care that would have happened regardless of the program. So they are inexpensive to obtain. They also carry a credibility that no self-reported number can match, because no one at the program chose, ordered, performed, or recorded the tests. The exchange returned 8.77 million relevant observations on program members.

2.2 Cohort and selection

Of 417,307 members with a dispensed GLP-1 and a record link, 49,896 (12.0%) had at least one measure paired across the first fill. Of those, 33,488 also had a matched weight. That 12.0% paired-ascertainment fraction is itself a reportable quantity. It is the share of a telehealth cohort for whom outside clinicians happened to record the same measure both before and after initiation. It sets the current ceiling on how much of such a cohort can be followed on third-party-measured outcomes. Members of this paired-outcomes cohort averaged 46.6 years of age. Of them, 87.2% were female, and mean body mass index was 30.7. The median member had 14 dispensed fills. At the first fill, 57.8% received tirzepatide and 42.2% semaglutide.

2.3 Measures, mapping, and cleaning

Observations were mapped from LOINC codes to ten measures. Duplicate records returned by more than one exchange source were collapsed. Same-day repeats were averaged. Values outside plausibility ranges fixed before the analysis were dropped, which removed 0.11% of the 8.77 million readings. Disease-group membership depends only on an outside diagnosis dated before the first fill. It never depends on the baseline value of the measure subsequently reported.

2.4 Measurement windows

Baseline is the reading nearest the first dispensed fill, taken anywhere from 365 days before it to 14 days after. Follow-up is the latest reading from 60 to 1,000 days after. So no member contributes a follow-up value from the first two months of treatment. Median follow-up was 325 days. The 14-day allowance on the baseline side means a small number of baseline readings fall just after the member joined. Two weeks is short relative to the time these measures take to move. If anything it understates the change, by letting early treatment appear in the baseline.

3 Statistical methods

3.1 Causal framework

Figure 1 states the identifying structure the decomposition relies on. Program initiation affects the biomarkers through two paths. The mediated path runs through weight loss. The direct path gathers everything else the program does: the medication's non-weight effects together with the clinical review, titration, nutrition support, and monitoring. The decomposition separates the mediated path from the direct one. It does not divide the direct path into the drug and the care around it. One unobserved structure is drawn as a dashed node. We leave it inside the beyond-weight term rather than claim to remove it. It is an engagement or health-behavior factor that selects who enters and who keeps testing. We treat it as residual confounding and quantify robustness where we can, rather than claim to have adjusted it away. We do not claim the beyond-weight term is causal. We claim it is the best observational separation of weight from the rest of the program available under these assumptions.

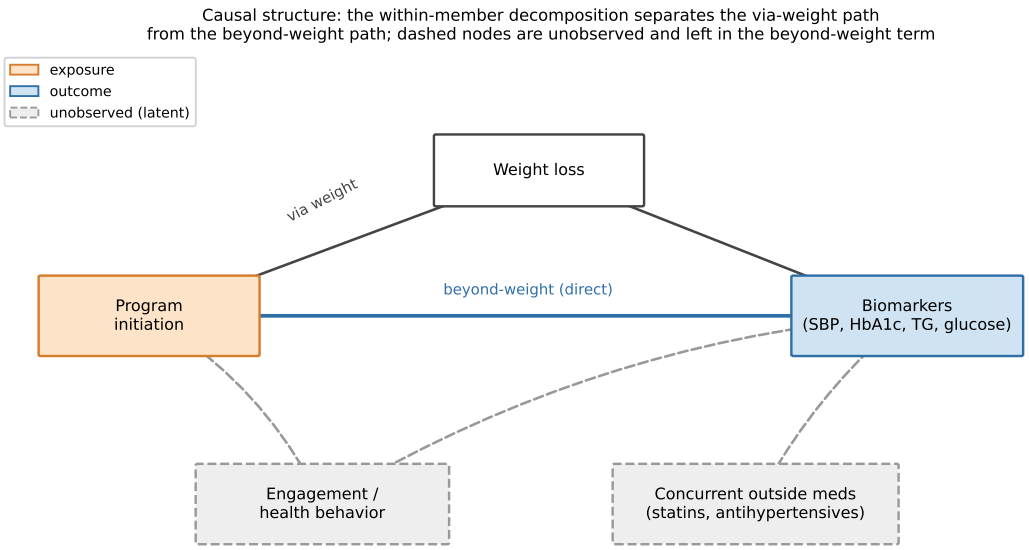


Figure 1: Causal structure for the decomposition. The mediated (via-weight) path and the direct (beyond-weight) path both run from program initiation to the biomarkers; the within-member models separate them. The dashed node is unobserved and remains inside the beyond-weight term.

3.2 Primary paired changes

For each measure we compare each member's baseline and follow-up reading. We test the paired difference with a paired t-test and a Wilcoxon signed-rank test. We report changes separately for members who were out of range at baseline and for the full diagnosed group. Averaging a member who came down 24 points with a member who had nothing to correct describes neither.

3.3 Dose-response

We sort weight-matched members into quartiles of percent body weight lost. We test the trend in each measure across quartiles, after verifying that baseline values are balanced across quartiles. Balance is the precondition for reading the gradient as a response to weight lost.

3.4 Within-member decomposition

The decomposition rests on two within-member models. Each is fit with a member-specific intercept, so that everything fixed about a member is absorbed: their genetics, baseline health, which clinician they see, and how often they are tested. Standard errors are clustered by member throughout.

The first model estimates how a member's own laboratory value moves with their own weight. It uses only post-initiation readings, so that none of the slope comes from the act of joining:

$$y_{it} \;=\; \alpha_i \;+\; \beta\, w_{it} \;+\; \gamma\, t \;+\; \varepsilon_{it}, \qquad t \geq \text{initiation}, \tag{1}$$

where y_{it} is the laboratory value, w_{it} the contemporaneous weight, α_i the member intercept, and t a linear time term that absorbs drift. The coefficient β is the per-unit-weight slope, identified

entirely within member. The weight-mediated part of the improvement is $\beta\,\overline{\Delta w}$, the slope applied to the mean weight change of 24.2 pounds.

The second model estimates the total step at initiation. It uses every reading from three years before to two years after the first fill, and it needs no weight reading. That matters because member weights begin only when the first prescription is dispensed:

$$y_{it} \;=\; \alpha_i \;+\; \gamma\, t \;+\; \delta\, D_{it} \;+\; \varepsilon_{it}, \tag{2}$$

where $D_{it} = 1$ for readings after initiation. The coefficient δ is the total step, net of the member intercept and the linear trend. The part of the improvement beyond weight is then $\delta - \beta\,\overline{\Delta w}$, and the weight share is

$$\text{weight share} \;=\; \frac{\beta\,\overline{\Delta w}}{\delta}. \tag{3}$$

The step in (2) is identified from the break in a member's own trajectory against their own history and a linear trend. So it is not driven by the level of the baseline. That is the sense in which the decomposition is more robust than the raw paired change. As a robustness check we refit (2) after deleting every reading in the 90 and 180 days before initiation, in case readings taken close to enrollment were unusual.

The weight share is a residual: $\beta\,\overline{\Delta w}$ divided by the step. So it is only as good as β, and β is pulled in both directions here. Three forces bias it toward zero, which would understate weight's share. First, weight loss is close to linear in time for the first several months. So the weight path w_{it} and the linear time term t in (1) are nearly collinear, and the time term absorbs part of weight's effect. Second, the laboratory response lags and accumulates while β is estimated contemporaneously, so a slow response reads as a smaller slope. Third, outside laboratory dates must be matched to program weight dates with error, which attenuates a contemporaneous slope further. Two forces push the other way, which would overstate weight's share. Within a member, weight falls as the dose is escalated, so post-initiation weight proxies for dose and for the drug's direct, non-weight effects, and β absorbs some of those. And the collinearity runs both ways, so if the direct effect ramps nonlinearly while weight falls, weight can steal from the time term rather than the reverse. The exposure adjustment in Table 3 does not settle this. It is fit on the cross-member gradient model, not on the within-member model (1) the decomposition uses. We therefore read the weight share as likely mismeasured but of uncertain direction, not as a bound. We treat the lagged and nonlinear specifications that would resolve it as work still to do. One caveat belongs with the mediator itself. The weights are captured by the program rather than by the outside clinicians. So unlike the outcomes, the mediator does not carry the third-party credibility the rest of the design rests on.

3.5 Within-member event study

Separately from the decomposition, we fit each member an ordinary-least-squares slope in the period before initiation and another after it, requiring at least two readings in each period. We compare the annualized pre- and post-initiation rates by paired t-test. We report this for every measure for which a pre and a post slope can be formed, rather than keeping only the ones that move, so that the test is not circular. A measure whose rate of improvement steps down at initiation is behaving as a treatment response. We carry only those forward into the claims above. The measures that do not step down are shown beside them.

3.6 Software

Analyses were performed in Python 3.12 with pandas, numpy, scipy, and statsmodels. Figures were made in matplotlib. The within-member models were fit by ordinary least squares with member fixed effects and standard errors clustered by member. The matched-difference intervals were fit by bootstrap. The aggregate tables and the reproducing notebook are in the data appendix.

4 Results

4.1 Primary outcomes across the diagnosed groups

The change across each full diagnosed group, out of range or not, is reported in Table 1. The group is defined by a prior diagnosis rather than by the value we subsequently measure. This is not a clean treatment effect. Secular trend and the heavier surveillance of sicker patients are both still in it. There is no untreated comparison group. With that said, systolic blood pressure fell 4.85 mmHg across 16,008 members with hypertension and 4.27 mmHg across 3,908 with established cardiovascular disease. HbA1c fell 0.57 points in type 2 diabetes and 0.35 in prediabetes. LDL fell 8.55 mg/dL and triglycerides 26.98 mg/dL in dyslipidemia. Each row is a different group measured on a different scale, so the rows are not comparable with one another. The type 2 diabetes group is small and starts at a low 6.05%, which indicates a largely controlled population.

Table 1: Paired changes across the full diagnosed group. They are not counterfactual effect estimates. Before and after are rounded for display, so each change is computed from the unrounded values.

Group	Measure	Before	After	Change	n
Hypertension	Systolic BP (mmHg)	128.8	124.0	−4.85	16,008
Existing CVD	Systolic BP (mmHg)	125.2	121.0	−4.27	3,908
Type 2 diabetes	HbA1c (%)	6.05	5.48	−0.57	953
Prediabetes	HbA1c (%)	5.67	5.32	−0.35	4,891
High cholesterol	LDL-C (mg/dL)	116.8	108.3	−8.55	8,294
High cholesterol	Triglycerides (mg/dL)	135	108	−26.98	8,296

4.2 The out-of-range subgroup

A larger set of numbers circulates for programs like this one, drawn from the subgroup that was out of range at baseline. We report them for completeness (Table 2). Among members out of range at baseline, systolic blood pressure fell 13.4 mmHg (from 141.0 to 127.7), HbA1c fell 1.46 points (from 7.42 to 5.97), and LDL cholesterol fell 24.2 mg/dL (from 155.7 to 131.6). Most finished inside the range a clinician treats to.

4.3 Improvement rises with weight lost

We sort weight-matched members (n = 33,488) into quartiles of percent body weight lost. The improvement rises monotonically with weight lost in every measure tested, from baselines that are balanced across quartiles (Figure 2). In the hypertension group the four quartiles lost 0.7%, 7.0%, 13.0%, and 23.8% of body weight, from nearly identical starting pressures of 128.1 to 129.9 mmHg. Their systolic reductions ran 2.5, 4.3, 6.4, and 8.0 mmHg. The per-point-of-weight-lost slopes are

Table 2: Paired changes in the subgroup out of range at baseline.

Condition	Measure	Before	After	Change	% in range after
Hypertension	Systolic BP (mmHg)	141.0	127.7	−13.4	58%
Hyperglycemia	HbA1c (%)	7.42	5.97	−1.46	80%
Hyperlipidemia	LDL-C (mg/dL)	155.7	131.6	−24.2	47%

Out-of-range thresholds: 130 mmHg, 6.5% HbA1c, 130 mg/dL LDL. Group sizes 7,267, 214, and 2,940. The 80% here is the share of out-of-range diabetic members finishing below 6.5%, which is distinct from the prediabetes-to-normal figure in Section 5.

−0.227 mmHg for systolic pressure ($p < 10^{-39}$), −0.014 points for HbA1c ($p < 10^{-78}$), and −1.47 mg/dL for triglycerides ($p < 10^{-48}$). If the changes were an artifact of repeated measurement or of simply being enrolled, there is no reason they should track the amount of weight lost. They do. This is consistent with a weight effect, but it does not prove one.

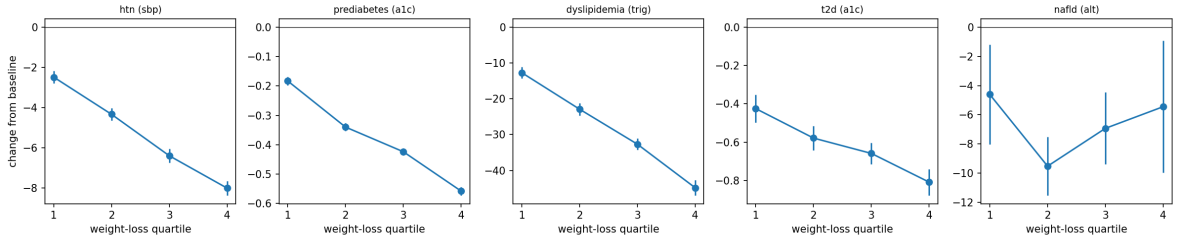


Figure 2: Improvement rises with weight lost, from balanced baselines. Weight-matched members, $n = 33{,}488$; points are mean change with 95% confidence intervals; quartiles are ranks, with mean body weight lost 0.7%, 7.0%, 13.0%, and 23.8%.

4.4 The gradient is not a dose effect in disguise

Members who lose more weight also take more drug for longer, so the gradient could be a dose effect. We added the number of dispensed fills, which agent the member received, and the length of follow-up to the per-weight model. The slopes were essentially unchanged: the systolic slope moved 2.9%, HbA1c 2.3%, and triglycerides 0.5% (Table 3). This addresses the recorded differences between members. It cannot address the unrecorded ones.

Table 3: Per-percent-weight slopes before and after adjustment for medication exposure.

Measure	Crude slope	Adjusted slope	% change	n
Systolic BP (mmHg)	−0.239	−0.246	+2.9	30,254
HbA1c (%)	−0.0122	−0.0119	−2.3	7,276
Triglycerides (mg/dL)	−1.262	−1.268	+0.5	9,671

Slopes are change per one percent of body weight lost. Adjustment adds fill count, agent, and follow-up length. All $p < 10^{-60}$ before and after.

4.5 The decomposition: weight's share, and why its direction is uncertain

The within-member decomposition attributes 42% of the systolic step to weight, 46% of HbA1c, 39% of triglycerides, and 28% of fasting glucose (Table 4, Figure 3), with the remainder arriving beyond weight. These shares inherit the two-sided attenuation of β described in the methods. So we read them as uncertain in direction rather than as bounds, and we do not draw a minority-versus-majority conclusion from point estimates this fragile. On blood pressure the measured split is 1.40 mmHg through weight against 1.98 mmHg through everything else. The weight slopes that carry the mediated part are precisely estimated and pass a sign check, though their size still inherits the collinearity between the weight path and the time term described in the methods, so we do not call them clean. Per 10 pounds, systolic pressure moves 0.58 mmHg, HbA1c 0.043 points, triglycerides 2.94 mg/dL, and glucose 0.57 mg/dL, all estimated from post-initiation readings. HDL cholesterol moves in the opposite direction as weight falls, which is the sign check a generic data bias would fail. Deleting readings in the 90 and 180 days before initiation shrinks the step by 3% on glucose, 10% on systolic pressure, 18% on HbA1c, and 19% on triglycerides, and it stays large and significant throughout. The figures reported use the full record. Both components are significant on every measure, all $p < 10^{-8}$.

Table 4: Within-member decomposition of each improvement into a weight-mediated part and a part beyond weight.

Measure	Total step	Via weight	Beyond weight	Weight share
Systolic BP (mmHg)	−3.38	−1.40	−1.98	42% (37–46)
HbA1c (points)	−0.230	−0.105	−0.125	46% (36–55)
Triglycerides (mg/dL)	−18.3	−7.1	−11.2	39% (23–55)
Fasting glucose (mg/dL)	−4.95	−1.37	−3.58	28% (19–37)

Step from (2); weight-mediated part is $\beta\,\overline{\Delta w}$ from (1) at the mean 24.2 pounds lost. Reading counts: systolic 896,768 from 60,333 members; HbA1c 56,729 from 16,214; triglycerides 75,145 from 21,323; glucose 187,077 from 30,247. Ranges are the 95% sampling interval on the share. They do not include the specification and collinearity uncertainty discussed in the methods, which is larger and not quantified here. The within-member weight slope β is attenuated in both directions (see text), so these shares are uncertain in direction, not bounds. Fasting glucose and triglycerides are exploratory, because they do not survive the multiplicity correction in Section 4.

4.6 Reading the decomposition next to SELECT

Our decomposition should be read next to the trial that already addressed the weight question for hard outcomes. In SELECT, semaglutide cut major adverse cardiovascular events in people with obesity and established cardiovascular disease. The prespecified analysis found that early in-trial weight loss was not a significant mediator of that benefit. Waist-circumference reduction accounted for up to about a third of it, and measured risk factors jointly for about a third. That left most of the effect unexplained by the adiposity and risk-factor changes that were measured [Lincoff et al., 2023, [author list to verify], 2025]. We do not measure hard events and cannot restate that result. What we can do is note the direction. SELECT established a weight-independent component of GLP-1 benefit for hard events. Our surrogate decomposition is consistent in sign, with a substantial part of each biomarker change arriving beyond weight. We do not set our surrogate share next to SELECT's hard-event mediation as a quantitative contrast. The two differ in endpoint, population, and estimation method at once, and our share is uncertain in direction, so a numeric comparison would read more into the pair than it can hold. The qualitative agreement

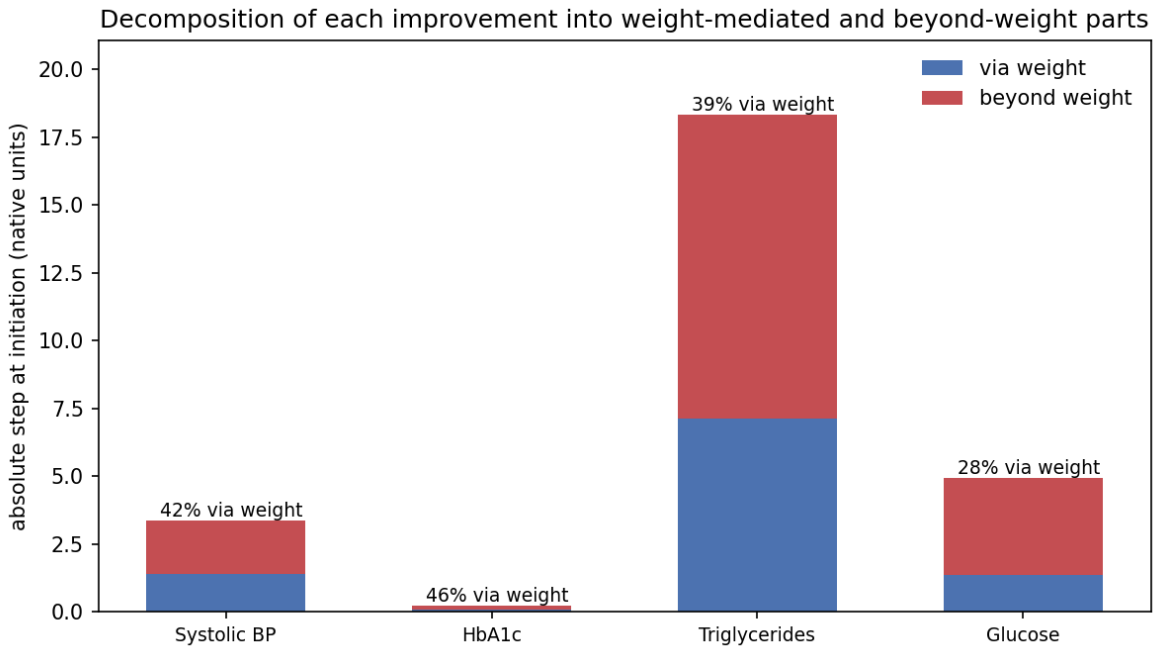


Figure 3: The measured weight-mediated share of each improvement, with the remainder arriving beyond weight. The within-member weight slope is attenuated in both directions, so these shares are uncertain in direction rather than bounds.

in sign is the most the two support. Our contribution is to measure the biomarker figure in a real-world population that SELECT's cardiovascular-disease enrollment criteria exclude. Because our endpoints are surrogates, this is weaker evidence than SELECT on what finally matters. We do not present it as a replication of the cardiovascular result.

4.7 Within-member event study across all measures

Table 5 reports the annualized rate of change before and after initiation for every measure for which both slopes can be formed. The measures that do not respond sit beside the ones that do. Five step down at initiation: systolic and diastolic blood pressure, HbA1c, fasting glucose, and triglycerides. Total cholesterol does the opposite, decelerating from -9.1 to -3.7 mg/dL per year. LDL shows no significant change in rate ($p = 0.07$). ALT does not move at all ($p = 0.86$). We carry only the five that accelerate into the claims above. Multiplicity matters here, because this is a family of eight tests. Systolic and diastolic blood pressure and HbA1c survive a Holm correction with a large margin. Fasting glucose ($p = 0.04$) and triglycerides ($p = 0.05$) do not. So we treat those two as exploratory rather than confirmed, here and in the decomposition. We make no claim about cholesterol.

4.8 Replication

The entire pipeline was re-extracted from source on 15 October 2026 at roughly four times the earlier sample. Every condition-group result held its direction and magnitude.

Table 5: Within-member annualized rate of change, before and after initiation, for every measure with a pre- and post-initiation slope.

Measure	n	Pre /yr	Post /yr	p	Pattern
Systolic BP (mmHg)	31,527	−1.24	−3.47	$< 10^{-29}$	accelerates
Diastolic BP (mmHg)	31,516	−0.47	−1.91	$< 10^{-24}$	accelerates
HbA1c (%)	2,098	−0.09	−0.20	$< 10^{-6}$	accelerates
Fasting glucose (mg/dL)	7,630	−0.76	−1.85	0.04	accelerates
Triglycerides (mg/dL)	2,952	−8.24	−13.13	0.05	accelerates
Total cholesterol (mg/dL)	2,957	−9.11	−3.74	$< 10^{-3}$	decelerates
LDL-C (mg/dL)	2,935	−7.34	−5.05	0.07	no change
ALT (U/L)	4,785	−2.23	−2.46	0.86	no change

Annualized ordinary-least-squares slopes fit per member before and after the first fill, compared by paired t-test. We carry forward only the measures that accelerate; the rest are shown rather than dropped silently.

5 Clinical significance of changes of this size

A laboratory value improving and a person becoming healthier are different claims. The second needs evidence from trials that followed comparable people for years while counting events. That evidence was not generated here. The figures in this section are projections from published trials applied to the changes we measured, not events observed at Mochi. They assume the improvements are sustained rather than held for the eleven months we observe (Table 6). We include them because they are the correct way to read the size of a change, and we keep them strictly separate from our measurements. We scale the full-population changes as the central projection, and show in parentheses what the larger out-of-range changes would give. Those parenthetical numbers are illustrative ceilings, not central estimates. The projections are the clinical weight a change of that size carries if produced, not an incremental effect of the program.

Two cautions attach specifically to this cohort. A single later value below a threshold is not sustained remission. HbA1c also reflects red-cell lifespan as well as blood sugar, so iron deficiency raises it [Kim et al., 2010] and ethnicity shifts it [Likhari and Gama, 2010]. With a membership that is 87% female and in midlife, iron-deficiency effects in particular are plausible here. Because iron status was not measured, its contribution is unknown and we do not estimate it.

6 What we deliberately did not analyze, and why

Four questions sit close to this one and are kept out on purpose. First, we do not split the beyond-weight term into the drug and the care around it. That split needs variation in who receives which component, which this single arm does not contain, and reporting a guess would misrepresent what the design can see. Second, we do not estimate clinical events prevented, years of life gained, or cost avoided. The data are laboratory values, not events, and the trial projections in Section 5 are kept explicitly separate for that reason. Third, we do not analyze safety, discontinuation, or tolerability. Those need adverse-event ascertainment and a different denominator than outcome pairing provides. Fourth, we do not separate branded from compounded formulation, which is a distinct question with its own confounding and belongs in its own study. Mixing any of these into the present analysis would produce estimates that mean neither thing clearly.

Table 6: What the published literature attaches to changes of the size measured here. Projections, not events counted at Mochi.

Change: full population (out-of-range)	Scaled from	Associated with the full-population change, if sustained
Systolic BP −4.85 mmHg (−13.4)	Ettehad et al. [2016]; Blood Pressure Lowering Treatment Trialists' Collaboration [2021]	≈10% fewer major cardiovascular events, 13% fewer strokes, 6% lower all-cause mortality (out-of-range ceiling ≈26%, 34%, 17%)
LDL-C −8.55 mg/dL (−24.2)	Cholesterol Treatment Trialists' Collaboration [2010]	≈5% fewer major vascular events per year of treatment (out-of-range ceiling ≈14%)
HbA1c −0.57 points (−1.46)	Stratton et al. [2000]	≈21% fewer microvascular complications, 8% fewer myocardial infarctions, though at this group's low 6.05% baseline even that is doubtful [Kianmehr et al., 2022]
Prediabetes to normal, 80% of 2,365	Gong et al. [2019]; Vazquez Arreola et al. [2026]	hazard ratio 0.41 for cardiovascular death or heart-failure hospitalization

The blood-pressure relationship is log-linear, so the 13.4 mmHg figure scales from the per-10-mmHg estimates. Multifactorial treatment of the cluster at once gained 7.9 years of life over 21 years in Steno-2 [Gæde et al., 2016]. Counter-evidence belongs here too: intensive lifestyle weight loss did not reduce cardiovascular events in established diabetes [The Look AHEAD Research Group, 2013], lowering triglycerides 26% did not reduce events in one trial [Das Pradhan et al., 2022], and at the low HbA1c baseline of this cohort no life-expectancy gain is projected from moving within range [Kianmehr et al., 2022].

7 Limitations

There is no untreated comparison group. So the paired changes are not effect estimates against a counterfactual, and only the within-member gradient and the trajectory break carry inferential weight. The program term in the decomposition bundles the medication with the clinical review, titration, nutrition support, and monitoring. The analysis separates weight from that bundle but cannot divide the bundle itself. So the defensible claim is that a substantial part of the measurable improvement does not travel through weight, not that the clinical care produces it. The live alternative is an engagement confounder, that members who lose more weight also manage the rest of their health more reliably. The counter is that such an effect would have to scale with pounds lost across several organ systems at once to reproduce the gradients, which is possible but not the simplest account. The measures are laboratory values rather than clinical events, so no count of events prevented, years gained, or cost avoided follows from them. Safety and discontinuation were not in scope.

8 Interpretation

The first thing to take from this is not a number but a measurement regime. The cardiometabolic changes reported here were recorded by the members' own outside clinicians, retrieved through nationwide health-information exchange, with no involvement from the program that treated them. That is a standard of outcome verification that direct-to-consumer telehealth has not previously met. On those outcomes, the measurements improved. The improvements scale with weight lost from balanced baselines. They survive adjustment for drug exposure. For the measures we keep, they appear as an acceleration at initiation in member-level trajectories that were flat or improving only slowly before. Initiation also coincides with a broader health-activation episode, a new clinician visit and concurrent prescriptions among them, so this single arm cannot assign that step to the drug alone. The decomposition then finds a substantial part of each improvement arriving beyond weight, with the exact share uncertain because the within-member weight slope is attenuated in both directions. That is consistent in sign with the weight-independence SELECT established for hard events. The simplest reading of that beyond-weight part is the GLP-1's direct, non-weight pharmacology, which is well documented for this drug class. The clinical care wrapped around the prescription, the review, titration, nutrition support, and monitoring, is one of several other components sitting inside the same term. This single arm cannot separate the drug from the care. So we do not claim that the care rather than the drug produces the beyond-weight share. What the measurement regime does establish is narrower and still useful. A program that dispenses these molecules and measures only weight is seeing a fraction of what its members' own clinicians record, and that fuller picture is now observable. Separating the drug from the care is the most valuable study to run next.

Appendix: registered-dietitian sub-study

This sub-study uses a separate, non-overlapping cohort and answers a different question from the main analysis. So it is reported apart from it, and its outcomes come from program records rather than from outside clinicians. The cohort is members with a GLP-1 fill between January 2023 and August 2025. The exposure is a registered-dietitian visit in the first 180 days. The outcomes are twelve-month weight loss and twelve-month persistence. Members who choose a dietitian differ from those who do not. So we matched one to one on a propensity score built from sex, age, body mass index, weight, and six recorded conditions, with a 0.02 caliper. We summarize the persistence result with an E-value [VanderWeele and Ding, 2017]. Of 241,482 members, 66,432 (27.5%) saw a dietitian within 180 days. After matching, a dietitian visit was associated with 1.94 percentage points more twelve-month weight loss (from 12.5 to 14.4%, 95% CI 1.79 to 2.09, 28,768 pairs), against an unadjusted gap of 1.89 points. It was also associated with 6.32 points more twelve-month persistence (from 38.4 to 44.7%, 95% CI 5.81 to 6.83, 66,431 pairs, risk ratio 1.17), against an unadjusted gap of 7.73. The persistence result carries an E-value of 1.60, so a moderate unmeasured confounder could overturn it. It is suggestive rather than decisive.

Data appendix

Every number in this paper reproduces from the de-identified aggregate tables in `data/`: the primary and full-population paired changes (`table_primary_out_of_range.csv`, `table_full_population.csv`), the dose-response quartiles (`disease_tableB3_quartiles.csv`), the exposure-adjusted slopes (`table7_exposure_a`) the pre/post rates (`table8b_pre_post_rates.csv`), the decomposition (`table9_weight_decomposition.csv`),

the cohort descriptors (`cohort_descriptors.json`), and the nutrition results (`nutrition_results.json`). The notebook `notebooks/01_reproduce_decomposition.ipynb` regenerates the tables and figures. Patient-level files are not distributed.

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